

VibeCoding Lab 4 — Chat / LLM History

Pygame Skyscraper Stack

Original repo: SETAPESU26/54_skyscraper_stack

Student fork: gaganaraj/54_skyscraper_stack

Purpose

This PDF records the available conversation history and the prompts/instructions used during the VibeCoding Lab 4 workflow. It is a structured record of the work completed in this conversation, rather than a platform-generated raw transcript with every UI message and timestamp.

1. Setup

- Set up the Windows project in C:\Users\lasus\VibeCoding_Lab4\54_skyscraper_stack.
- Created and activated a Python virtual environment.
- Installed Pygame 2.6.1.
- Opened the repository in VS Code and inspected game/block.py, game/game_engine.py, main.py and README.md.

2. Assignment Tasks

- Task 1: Fix the inverted overlap condition so $\text{overlap} > 0$ means successful placement.
- Task 2: Add perfect alignment, PERFECT feedback, bonus points and a three-perfect streak width expansion.
- Task 3: Turn trimmed overhangs into falling and rotating debris affected by gravity.
- Task 4: Add atmospheric background progression through twilight, dusk, night/starfield and stratosphere stages.
- Submission requirements: BEFORE video, AFTER video, updated code and Chat/LLM history.

3. Task 1 — Overlap Fix

- Original logic used $\text{overlap} \leq 0$ for a successful drop.
- Changed it to $\text{overlap} > 0$.
- Test confirmed overlapping blocks land successfully and non-overlapping blocks trigger game over.
- Recorded the BEFORE video showing the original broken behavior.

4. Task 2 — Perfect Alignment

- Added `bonus_score`, `perfect_streak`, `perfect_popup_timer` and PERFECT display.
- Implemented a practical alignment tolerance of 10 pixels for the demonstration.
- Perfect placement keeps the width, snaps the x-position, awards bonus points and increments the streak.
- Three consecutive perfect placements slightly expand the block width, up to its base width.
- Normal placement trims to overlap and resets the streak.
- User tested the feature successfully.

5. Task 3 — Falling Debris

- Added a Debris object with horizontal velocity, vertical velocity, gravity, rotation and angular velocity.
- Created debris from left/right overhangs when a block is trimmed.
- Updated debris every frame and removed it after it leaves the screen.
- User tested the feature and observed falling/rotating debris.

6. Task 4 — Atmospheric Background

- Added vertical atmospheric gradients controlled by tower height.
- Stages include twilight blue, dusk purple, night with stars, and stratosphere black.
- Added deterministic star positions for the night stage.
- User tested the feature and observed the dark night background and visible stars.

7. GitHub / Fork / Push

- Initial push to SETAPESU26/54_skyscraper_stack failed with GitHub 403 permission denied.
- Created a personal fork under gaganaraj.
- Changed origin to https://github.com/gaganaraj/54_skyscraper_stack.git.
- Verified the remote with `git remote -v`.
- Successfully ran `git push -u origin main`.
- GitHub confirmed main -> main and configured origin/main tracking.

8. Videos and Submission

- BEFORE video: completed.
- AFTER video: completed and uploaded as AFTER.mp4.
- Updated code: successfully pushed to the personal GitHub fork.
- Chat/LLM history: this PDF.

9. Git History Note

- The completed project was pushed successfully. Tasks 1–3 were not preserved as three separate commits; their changes were bundled into the commit used for the Task 3 stage. Task 4 and supporting changes were committed separately. This document does not fabricate separate task commits.